

Partial Differential Equations

Problem Sheet 3

1. Solve

(a)

$$\begin{cases} u_x + xu_y = u, \\ u(1, y) = h(y). \end{cases}$$

(b)

$$\begin{cases} yu_x + xu_y = 0, \\ u(x, 0) = x^4. \end{cases}$$

2. Consider the following semilinear equation

$$\begin{cases} u_x + bu_y = u^2, \\ u(x, 0) = \cos(x). \end{cases}$$

Find the largest neighbourhood of the x -axis where the above PDE admits a solution.

3. Solve the Cauchy problem

$$\begin{cases} xu_x - yu_y = u - y, & x > 0, y > 0, \\ u(y^2, y) = y, & y > 0. \end{cases}$$

Verify the transversality condition along the initial curve. Draw the initial curve and the projected characteristics intersecting the initial curve. *

4. Find the solution of the linear equation $u_x + xu_y = y$ satisfying the initial condition $u(0, y) = g(y), g \in \mathbb{R}$. Draw the projected characteristics intersecting the initial curve

$$\Gamma = \{(0, s) : s \in \mathbb{R}\}.$$

5. Solve the initial value problem

$$\begin{cases} xu_x + yu_y + u_z = u, \\ u(x, y, 0) = \sin(xy), \end{cases}$$

where $u = u(x, y, z)$.

6. Solve

$$u_y = xu u_x$$

subject to

$$u(x, 0) = x, \quad x \in \mathbb{R}.$$

(You may get only an implicit solution.)

7. Consider the equation

$$yu_x - xu_y = 0, \quad y > 0.$$

Check for each of the following initial conditions whether the problem is solvable. If it is solvable, find a solution. If it is not, explain why.

(a) $u(x, 0) = x^2$.

(b) $u(x, 0) = x$.

(c) $u(x, 0) = x, \quad x > 0$.

8. Let $u = u(x, t)$ be the solution of the following initial value problem:

$$\begin{cases} u_t + u_x = 0, & x \in \mathbb{R}, t > 0, \\ u(x, 0) = u_0(x), & x \in \mathbb{R}, \end{cases}$$

where $u_0 : \mathbb{R} \rightarrow \mathbb{R}$ is an arbitrary C^1 function. Prove or disprove the following two statements.

S_1 : If $A_t := \{x \in \mathbb{R} : u(x, t) < 1\}$ and $|A_t|$ denotes the Lebesgue measure of A_t for every $t \geq 0$, then $|A_t| = |A_0|$, $\forall t > 0$.

S_2 : If u_0 is Lebesgue integrable, then for every $t > 0$ the function $x \mapsto u(x, t)$ is Lebesgue integrable.

9. Consider the linear equation

$$au_x + bu_y = c(x, y)$$

with the initial condition $u(x, 0) = h(x)$. Assume a, b are constants and $b \neq 0$, and $c, h \in C^1$. Using the method of characteristics, derive the solution

$$u(x, y) = h(x - \gamma y) + \int_0^{y/b} c(a\tau + x - \gamma y, b\tau) d\tau, \quad \gamma = \frac{a}{b}.$$